Fall 2025

KAIST EE412

Foundation of Big Data Analytics

Midterm Solution

Name: _____

Student ID: _____

I agree to comply with the School of Electrical Engineering Honor Code.

Signature: _____

This exam is open notes, closed book, and closed computer. There are 24 pages in total. You have 150 minutes (1:00 – 3:30) to complete your exam. Please use this exam sheet to write answers. You may write in English or Korean.

Read the questions carefully and focus your answers on what has been asked. You are allowed to ask the instructor/TAs for help only in understanding the questions, in case you find them not completely clear. Be concise and precise in your answers and state clearly any assumption you may have made. Good luck!

Question	Score
1	/20
2	/15
3	/20
4	/20
5	/20
6	/25
Total	/120

1 (20 points) TRUE/FALSE

State if the following statements are true or false. Please write TRUE or FALSE in the space provided. Each problem is worth 1 point, but you will be penalized -1 point for an incorrect answer, so leave the answer blank if you don't know.

(a) In a distributed computing cluster, each node typically shares the same CPU and memory with other nodes on the rack.

Answer: FALSE

(b) If a Map Worker fails, all tasks it has completed must be redone, because its output is stored locally and not in the distributed file system.

Answer: TRUE

(c) In Spark, the GroupByKey transformation creates a wide dependency because each output partition may depend on multiple input partitions.

Answer: TRUE

(d) RDDs in Spark can be updated in place to modify existing records.

Answer: FALSE

(e) The *interest* of an association rule is always positive.

Answer: FALSE

(f) Market-basket analysis can be applied only to retail data and not to other domains such as student-club relationships.

Answer: FALSE

(g) The PCY algorithm can efficiently use the triangular-matrix method to count frequent pairs.

Answer: FALSE

(h) In the multistage algorithm, truly frequent pairs will always hash to frequent buckets across all stages.

Answer: TRUE

(i) The expected similarity between two minhash signatures equals the Euclidean distance between the original sets.

Answer: FALSE

(j) Longer minhash signatures lead to smaller expected errors when estimating similarity.

Answer: TRUE

(k) After minhashing, LSH guarantees that all pairs of similar documents will be identified.

Answer: FALSE

(l) In the banding technique, signatures are divided into several bands, and each band is hashed independently to identify candidate pairs.

Answer: TRUE

(m) A clustroid represents an existing data point that is closest to all other points within the same cluster.

Answer: TRUE

(n) Hierarchical clustering may stop merging clusters when the merged cluster's diameter exceeds a predefined threshold.

Answer: TRUE

(o) In the BFR algorithm, the discard set contains points that have not yet been assigned to any cluster.

Answer: FALSE

(p) In the GRGPF algorithm, a new point cannot be the clustroid immediately when it is added.

Answer: TRUE

(q) The eigenvectors of a symmetric matrix are always orthogonal to each other.

Answer: TRUE

(r) The first principal component corresponds to the eigenvector associated with the smallest eigenvalue of the covariance matrix.

Answer: FALSE

(s) The number of non-zero singular values in SVD equals the number of rows in the data matrix.

Answer: FALSE

(t) The low-rank approximation of a matrix obtained through SVD minimizes the reconstruction error among all matrices of the same rank.

Answer: TRUE

2 Finding Similar Items

Note: To receive full credit for each sub-question, **all entries must be correct** (no partial points will be given).

(a) [6 points] We want to create the characteristic matrix for given three documents $S1$, $S2$, and $S3$ using k-shingles with $k = 6$. We should normalize each document by converting to lowercase, and removing punctuation before forming 6-shingles.

- $S1$: Hanwha Eagles is the most popular sports club in Korea. To celebrate the playoff events, every seat is decorated with a single flower.
- $S2$: Hanwha Life Esports team always excites the crowds. Their events include tournaments against famous teams.
- $S3$: In October, Daejeon flower festival is held. The main events are flower exhibitions and marathon.

Construct the characteristic matrix by applying indexing function f, from which the following mapping is obtained:

$$\text{hanwha} \to 0, \quad \text{sports} \to 1, \quad \text{events} \to 2, \quad \text{flower} \to 3, \quad \text{crowds} \to 4.$$

Row	S_1	S_2	S_3
0: hanwha	1	1	0
1: sports	1	1	0
2: events	1	1	1
3: flower	1	0	1
4: crowds	0	1	0

(b) [4 points] Suppose we compute minhash signatures for documents as covered in class. Fill in the blanks of the signature matrix shown below using the following hash functions:

$$g_1(x) = 2x - 3 \pmod 5, \qquad g_2(x) = 2x + 3 \pmod 3,$$
$$g_3(x) = 4x + 2 \pmod 3, \qquad g_4(x) = 3x + 4 \pmod 5.$$

Hash fn	Sig(S_1)	Sig(S_2)	Sig(S_3)
g_1			
g_2			
g_3			
g_4			

Hash fn	Sig(S_1)	Sig(S_2)	Sig(S_3)
g_1	1	0	1
g_2	0	0	0
g_3	0	0	1
g_4	0	0	0

(c) [3 points] Based on the results in (a) and (b),

1. Compute the true Jaccard similarity for all pairs of sets.
2. Estimate the approximate Jaccard similarities using the minhash signatures.
3. Write any pair of sets whose true and estimated similarities are equal. If no such pair exists, write "none."

1. True Jaccard similarities.

$$S_1 = \{0, 1, 2, 3\}, \quad S_2 = \{0, 1, 2, 4\}, \quad S_3 = \{2, 3\}.$$

$$\mathrm{J}(S_1, S_2) = \frac{3}{5} = 0.60, \quad \mathrm{J}(S_1, S_3) = \frac{2}{4} = 0.50, \quad \mathrm{J}(S_2, S_3) = \frac{1}{5} = 0.20.$$

2. Estimated Jaccard from signatures (fraction of matching rows):

$$\hat{\mathrm{J}}(S_1, S_2) = \frac{3}{4} = 0.75, \quad \hat{\mathrm{J}}(S_1, S_3) = \frac{3}{4} = 0.75, \quad \hat{\mathrm{J}}(S_2, S_3) = \frac{2}{4} = 0.50.$$

3. Pairs with equal true and estimated similarities: **none**.

(d) [4 points] Let s_{12}, s_{13}, s_{23} be the true Jaccard similarities obtained from parts (c), where s_{ij} denotes the Jaccard similarity between i and j. Consider LSH banding configurations with $m = 4$ rows:

$$(r, b) \in \{(1, 4),\ (2, 2),\ (4, 1)\}$$

1. For each (r, b), write down the expressions for $P_{\text{cand}}(s_{12})$, $P_{\text{cand}}(s_{13})$, and $P_{\text{cand}}(s_{23})$ (numerical computation is not required).

(r, b)	$P_{\text{cand}}(s_{12})$	$P_{\text{cand}}(s_{13})$	$P_{\text{cand}}(s_{23})$
(1,4)			
(2,2)			
(4,1)			

2. Based on your results, simply explain how increasing r affects recall and false positive rate, and explain how increasing b affects recall and false positive rate.

Let $s_{12} = 0.60$, $s_{13} = 0.50$, $s_{23} = 0.20$. With $m = rb = 4$, the LSH candidate probability is $P_{\text{cand}}(s) = 1 - (1 - s^r)^b$.

(r, b)	$P_{\text{cand}}(s_{12})$	$P_{\text{cand}}(s_{13})$	$P_{\text{cand}}(s_{23})$
(1,4)	$1 - (1 - 0.60)^4$	$1 - (1 - 0.50)^4$	$1 - (1 - 0.20)^4$
(2,2)	$1 - (1 - 0.60^2)^2$	$1 - (1 - 0.50^2)^2$	$1 - (1 - 0.20^2)^2$
(4,1)	0.60^4	0.50^4	0.20^4

Effect summary:

- Increasing r (stricter AND within a band) decreases $P_{\text{cand}}(s)$ for all s: *lower recall, lower FPR.*
- Increasing b (more OR across bands) increases $P_{\text{cand}}(s)$ for all s: *higher recall, higher FPR.*

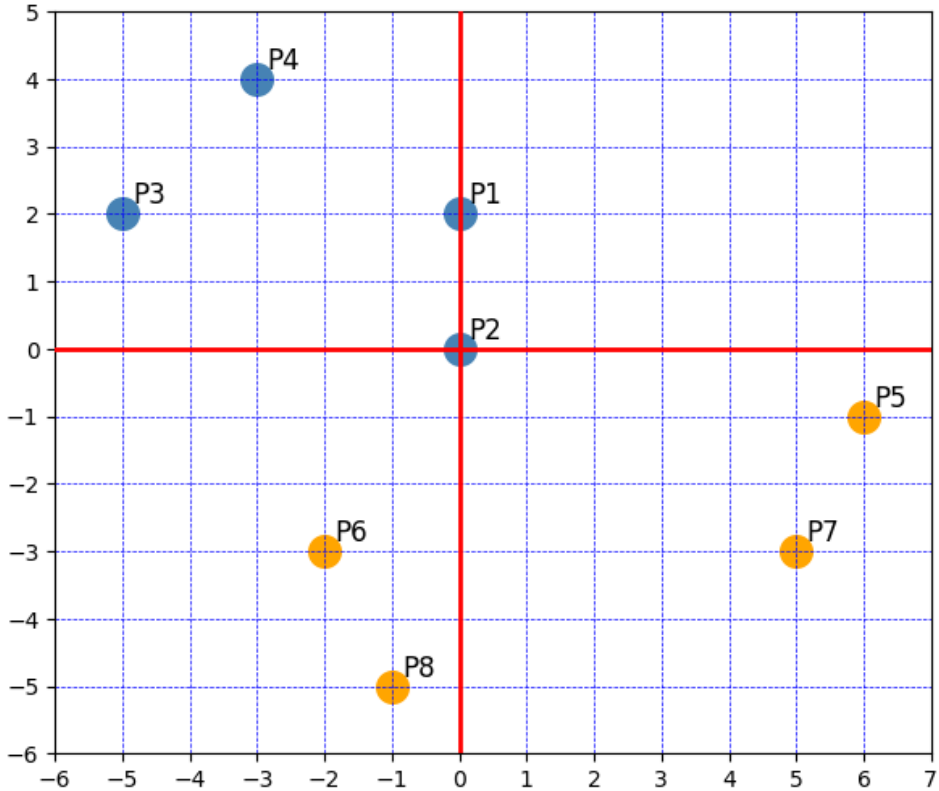
3 [20 points] Clustering

(a) [10 points] For this question, you will simulate the **CURE** algorithm. The data points are already clustered into two clusters $\mathbf{C_1}$ and $\mathbf{C_2}$. For each cluster, there are four data points as follows:

Cluster $\mathbf{C_1}$: $\mathbf{P_1} = (0, 2)$, $\mathbf{P_2} = (0, 0)$, $\mathbf{P_3} = (-5, 2)$, $\mathbf{P_4} = (-3, 4)$

Cluster $\mathbf{C_2}$: $\mathbf{P_5} = (6, -1)$, $\mathbf{P_6} = (-2, -3)$, $\mathbf{P_7} = (5, -3)$, $\mathbf{P_8} = (-1, -5)$

All distances use the **Euclidean (L2)** metric.



(i) [2 points] Find the coordinates for the centroid of each cluster using the given data points.

$\mathbf{C_1}$: _____

$\mathbf{C_2}$: _____

(i) Centroids.

$$\mathbf{C_1} : c_1 = \left(\frac{0+0-5-3}{4}, \frac{2+0+2+4}{4}\right) = (-2,\ 2),$$
$$\mathbf{C_2} : c_2 = \left(\frac{6-2+5-1}{4}, \frac{-1-3-3-5}{4}\right) = (2,\ -3).$$

(ii) [2 points] For each cluster, you will select 2 representative points. The first representative is the point farthest from the cluster centroid. The second representative is the point farthest from the first representative. Using the centroids calculated from the previous question, select 2 representative points for each cluster.

$\mathbf{C_1}$: The first rep. is _____ and the second rep. is _____

$\mathbf{C_2}$: The first rep. is _____ and the second rep. is _____

For $\mathbf{C_1}$ (centroid $c_1 = (-2, 2)$):

$$d(P_1, c_1) = 2, \quad d(P_2, c_1) = 2\sqrt{2} \approx 2.828,$$
$$d(P_3, c_1) = 3, \quad d(P_4, c_1) = \sqrt{5} \approx 2.236.$$

Farthest $\Rightarrow P_3 = (-5, 2)$. From P_3:

$$d(P_1, P_3) = 5, \quad d(P_2, P_3) = \sqrt{29} \approx 5.385, \quad d(P_4, P_3) = \sqrt{8} \approx 2.828.$$

Second rep $\Rightarrow P_2 = (0, 0)$.

$$\boxed{\text{Reps}(C_1) = \{P_3,\ P_2\}.}$$

For $\mathbf{C_2}$ (centroid $c_2 = (2, -3)$):

$$d(P_5, c_2) = \sqrt{20} \approx 4.472, \quad d(P_6, c_2) = 4,$$
$$d(P_7, c_2) = 3, \quad d(P_8, c_2) = \sqrt{13} \approx 3.606.$$

Farthest $\Rightarrow P_5 = (6, -1)$. From P_5:

$$d(P_6, P_5) = \sqrt{68} \approx 8.246, \quad d(P_7, P_5) = \sqrt{5} \approx 2.236, \quad d(P_8, P_5) = \sqrt{65} \approx 8.062.$$

Second rep $\Rightarrow P_6 = (-2, -3)$.

$$\boxed{\text{Reps}(C_2) = \{P_5,\ P_6\}.}$$

(iii) [3 points] Move each chosen representative 50% toward its cluster's centroid. Then, provide the final moved coordinates.

$\mathbf{C_1}$: _____ and _____

$\mathbf{C_2}$: _____ and _____

Shrink each representative by 50% toward the centroid.

$$p' = \frac{p+c}{2}.$$

$$C_1 : P_3' = (-3.5,\ 2), \quad P_2' = (-1,\ 1);$$
$$C_2 : P_5' = (4,\ -2), \quad P_6' = (0,\ -3).$$

(iv) [3 points] A new point $\mathbf{P_9} = (0, -1)$ is given. Using the **CURE** assignment rule, determine which cluster $\mathbf{P_9}$ should be assigned to. You must show your distance calculations to get full credit.

Assign $P_9 = (0, -1)$ **(nearest shrunk representative).**

$$\begin{aligned}
D(P_9, C_1) &= \min\{\|(0,-1) - (-3.5, 2)\|,\ \|(0,-1) - (-1,1)\|\} \\
&= \min\{\sqrt{3.5^2 + (-3)^2},\ \sqrt{1^2 + (-2)^2}\} = \min\{\sqrt{21.25},\ \sqrt{5}\} \\
&= \sqrt{5} \approx 2.236; \\
D(P_9, C_2) &= \min\{\|(0,-1) - (4,-2)\|,\ \|(0,-1) - (0,-3)\|\} \\
&= \min\{\sqrt{(-4)^2 + 1^2},\ \sqrt{0^2 + (-2)^2}\} = \min\{\sqrt{17},\ 2\} = 2.
\end{aligned}$$

$$\boxed{D(P_9, C_2) < D(P_9, C_1) \ \Rightarrow\ P_9 \text{ is assigned to } C_2.}$$

(b) [10 points] **A-CURE (Adjusted CURE)**
A-CURE is a variant of the CURE algorithm. Instead of moving all representative points by a fixed fraction toward the centroid, A-CURE adjusts the shrinkage rate for each representative based on its distance to the cluster centroid.

Let d_i be the Euclidean distance between a representative point P_i and the centroid c. The adjusted shrinkage ratio is

$$\alpha_i = \frac{1}{d_i},$$

and the moved coordinate is

$$P_i' = (1 - \alpha_i)P_i + \alpha_i\, c.$$

Use the same data points $\{P_1, \ldots, P_8\}$ and the Euclidean (L2) metric as in part (a).

(i) [4 points] Move each representative from part (a-ii) toward its cluster centroid using the adjusted shrinkage ratio α_i. Report the final coordinates for the two representatives of each cluster.

$\mathbf{C_1}$: _____ and _____

$\mathbf{C_2}$: _____ and _____

For C_1 (centroid $c_1 = (-2, 2)$, reps P_3, P_2):

$$d(P_3, c_1) = 3,\ \alpha = \tfrac{1}{3}, \quad P_3' = (1 - \tfrac{1}{3})(-5, 2) + \tfrac{1}{3}(-2, 2) = (-4,\ 2);$$
$$d(P_2, c_1) = \sqrt{8},\ \alpha = \tfrac{1}{\sqrt{8}} = \tfrac{1}{2\sqrt{2}}, \quad P_2' = \alpha(-2, 2) = \left(-\tfrac{1}{\sqrt{2}},\ \tfrac{1}{\sqrt{2}}\right).$$

For C_2 (centroid $c_2 = (2, -3)$, reps P_5, P_6):

$$d(P_5, c_2) = \sqrt{20} = 2\sqrt{5},\ \alpha = \tfrac{1}{2\sqrt{5}}, \quad P_5' = (6, -1) - \tfrac{1}{2\sqrt{5}}(4, 2) = \left(6 - \tfrac{2}{\sqrt{5}},\ -1 - \tfrac{1}{\sqrt{5}}\right);$$
$$d(P_6, c_2) = 4,\ \alpha = \tfrac{1}{4}, \quad P_6' = (-2, -3) + \tfrac{1}{4}(4, 0) = (-1,\ -3).$$

(ii) [6 points] The same new point as in part (a-iv), $P_9 = (0, -1)$, is given. Using the shrunk representatives for each cluster by A-CURE as in part (b-i), determine the cluster assignment for P_9. Show your intermediate steps to get full credit.

Assign $P_9 = (0, -1)$ **(nearest moved representative).**

$$\begin{aligned}
D(P_9, C_1) &= \min\{\|(0, -1) - (-4, 2)\|,\ \|(0, -1) - (-\tfrac{1}{\sqrt{2}},\ \tfrac{1}{\sqrt{2}})\|\} \\
&= \min\{\sqrt{16 + 9},\ \sqrt{\left(\tfrac{1}{\sqrt{2}}\right)^2 + \left(-1 - \tfrac{1}{\sqrt{2}}\right)^2}\} \\
&= \min\{5,\ \sqrt{\tfrac{1}{2} + \left(1 + \tfrac{1}{\sqrt{2}}\right)^2}\} = \min\{5,\ \sqrt{\tfrac{1}{2} + 1 + \sqrt{2} + \tfrac{1}{2}}\} \\
&= \min\{5,\ \sqrt{2 + \sqrt{2}}\} = \sqrt{2 + \sqrt{2}} \approx 1.848; \\
D(P_9, C_2) &= \min\left\{\left\|(0, -1) - \left(6 - \tfrac{2}{\sqrt{5}},\ -1 - \tfrac{1}{\sqrt{5}}\right)\right\|,\ \|(0, -1) - (-1, -3)\|\right\} \\
&= \min\left\{\sqrt{\left(-6 + \tfrac{2}{\sqrt{5}}\right)^2 + \left(\tfrac{1}{\sqrt{5}}\right)^2},\ \sqrt{1^2 + 2^2}\right\} \\
&= \min\left\{\sqrt{36 - \tfrac{24}{\sqrt{5}} + \tfrac{4}{5} + \tfrac{1}{5}},\ \sqrt{5}\right\} = \min\left\{\sqrt{36 - \tfrac{24}{\sqrt{5}} + 1},\ \sqrt{5}\right\} \\
&= \min\left\{\sqrt{37 - \tfrac{24}{\sqrt{5}}},\ \sqrt{5}\right\} = \sqrt{5} \approx 2.236.
\end{aligned}$$

$$\boxed{D(P_9, C_1) < D(P_9, C_2) \ \Rightarrow\ P_9 \text{ is assigned to } C_1 \text{ under A-CURE.}}$$

4 [20 points] Dimensionality Reduction

[10 points] Part A. There are n students and m clubs. Each student belongs to exactly one club. Let $X \in \mathbb{R}^{n\times m}$ be the *membership matrix* defined by

$$X_{ij} = \begin{cases} 1, & \text{if student } i \text{ is a member of club } j, \\ 0, & \text{otherwise.} \end{cases}$$

Hence, each row of X is a *one-hot* vector indicating the club to which the corresponding student belongs. Let n_j denote the number of students in club j, so that $\sum_{j=1}^{m} n_j = n$. Assume that each club has at least one member, i.e. $n_j > 0$ for all $j = 1, \ldots, m$. Consider the singular value decomposition (SVD) $X = U\Sigma V^\top$ where

$$U \in \mathbb{R}^{n\times m}, \quad V \in \mathbb{R}^{m\times m} \text{ is orthonormal}, \quad \Sigma = \operatorname{diag}(\sigma_1, \ldots, \sigma_m),$$

and the singular values satisfy $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_m \geq 0$. Here, $\operatorname{diag}(a_1, \ldots, a_m)$ denotes a diagonal matrix with diagonal entries $a_1, \ldots, a_m$ and zeros elsewhere.

(i) [2 points] Express $X^\top X$ in terms of $n_1, \ldots, n_m$.

$$(X^\top X)_{jk} = \sum_{i=1}^{n} X_{ij} X_{ik}.$$

Since each row of X is one-hot, each student belongs to only one club. Thus, for $j \neq k$, no student can be in both clubs, so $(X^\top X)_{jk} = 0$. When $j = k$, the sum counts the number of students in club j, giving $(X^\top X)_{jj} = n_j$. Therefore,

$$\boxed{X^\top X = \operatorname{diag}(n_1, n_2, \ldots, n_m)}.$$

(ii) [3 points] Express U, Σ, and V in terms of $n_1, \ldots, n_m$.

From part (a), $X^\top X = \operatorname{diag}(n_1, \ldots, n_m)$, whose eigenvalues are $n_1, \ldots, n_m$ and eigenvectors are the standard basis vectors. Hence,

$$V = I_m, \quad \Sigma = \operatorname{diag}(\sqrt{n_1}, \ldots, \sqrt{n_m}).$$

Since $X = U\Sigma V^\top = U\Sigma$, we have $U = X\Sigma^{-1}$, giving

$$U_{ij} = \begin{cases} \dfrac{1}{\sqrt{n_j}}, & \text{if student } i \text{ is in club } j, \\ 0, & \text{otherwise.} \end{cases}$$

Finally,

$$U^\top U = \Sigma^{-1} X^\top X \Sigma^{-1} = I_m,$$

so U has orthonormal columns.

$$\boxed{U_{ij} = \begin{cases} \dfrac{1}{\sqrt{n_j}}, & \text{if student } i \text{ is in club } j, \\ 0, & \text{otherwise,} \end{cases} \qquad \Sigma = \operatorname{diag}(\sqrt{n_1}, \ldots, \sqrt{n_m}), \quad V = I_m.}$$

(iii) [1 point] Let $n_{(1)} \geq n_{(2)} \geq \cdots \geq n_{(m)}$ denote the club sizes sorted in non-increasing order. When only the top $r < m$ singular values of X are retained (rank-r truncated SVD), express the *retained energy ratio* in terms of $n_{(1)}, \ldots, n_{(m)}$.

Since in the one-hot setting $\sigma_j^2 = n_j$, the retained energy when keeping the top r singular values and retained energy ratio is,

$$\sum_{j=1}^{r} \sigma_{(j)}^2 = \sum_{j=1}^{r} n_{(j)}, \qquad \boxed{R(r) = \frac{\sum_{j=1}^{r} n_{(j)}}{n}.}$$

(iv) [4 points] Let $\mathcal{C}_r$ denote the set of all matrices obtained by selecting **any** r **columns** of X and setting all other columns to zero (i.e., considering only r clubs). Let X_r denote the rank-r truncated SVD of X, i.e., the best rank-r approximation of X in the Frobenius norm. Determine whether there exists $Y^\star \in \mathcal{C}_r$ such that

$$\|X - Y^\star\|_F = \|X - X_r\|_F,$$

and explicitly characterize $Y^\star$ if it exists.

From part (a), $X^\top X = \mathrm{diag}(n_1, \ldots, n_m)$, so the columns of X are pairwise orthogonal with squared norms $\|X_{\cdot j}\|_2^2 = n_j$. From part (b), the SVD satisfies $V = I_m$, $\Sigma = \mathrm{diag}(\sqrt{n_1}, \ldots, \sqrt{n_m})$, and $U = X\Sigma^{-1}$. The rank-r truncation is

$$X_r \;=\; \sum_{j=1}^{r} u_{(j)} \sigma_{(j)} e_{(j)}^\top,$$

where $(\cdot)_{(j)}$ denotes ordering by $n_{(1)} \geq \cdots \geq n_{(m)}$, and $e_{(j)}$ are standard basis vectors. Since $u_j \sigma_j e_j^\top = \left(\frac{X_{\cdot j}}{\sqrt{n_j}}\right)\left(\sqrt{n_j}\right) e_j^\top = X_{\cdot j} e_j^\top$, we obtain

$$X_r \;=\; \sum_{j=1}^{r} X_{\cdot(j)} e_{(j)}^\top,$$

i.e., X_r equals X with all but the r largest columns zeroed. Therefore, if S indexes the r clubs with largest n_j, the matrix $Y^\star \in \mathcal{C}_r$ that keeps precisely those columns satisfies $Y^\star = X_r$ and

$$\|X - Y^\star\|_F \;=\; \|X - X_r\|_F.$$

Moreover,

$$\|X - Y^\star\|_F^2 = \sum_{j \notin S} \|X_{\cdot j}\|_2^2 = \sum_{j \notin S} n_j = \sum_{j=r+1}^{m} n_{(j)} = \sum_{j=r+1}^{m} \sigma_{(j)}^2 = \|X - X_r\|_F^2.$$

Thus, *truncated SVD is exactly equivalent to keeping the r largest clubs.*

[10 points] Part B. Now assume each student may join multiple clubs, so a row of X can contain more than one "1." Consequently, each row of X is a *multi-hot* vector representing all clubs to which the student belongs. Please answer the following questions.

(i) [2 points] Describe the meaning of each element $(X^\top X)_{jk}$ when students are allowed to join multiple clubs. Explain, in words, what this value tells us about the relationship between clubs j and k.

$$(X^\top X)_{jk} = \sum_{i=1}^{n} X_{ij} X_{ik} = \big|\{\, i \mid \text{student } i \text{ belongs to both clubs } j \text{ and } k \,\}\big|.$$

The diagonal entry $(X^\top X)_{jj} = n_j$ gives the total number of members in club j. Each off-diagonal entry $(X^\top X)_{jk}$ $(j \neq k)$ counts the number of students who belong to *both* clubs j and k. Therefore, $X^\top X$ describes co-membership between different clubs.

(ii) [3 points] Assume each club has exactly s members and that any two distinct clubs $j \neq k$ share exactly t members, where $0 < t \leq s$. Express the singular values of X in terms of s, t, and m.

Under the given assumptions, the club–club co-membership matrix is

$$X^\top X = \begin{bmatrix} s & t & \cdots & t \\ t & s & \cdots & t \\ \vdots & \vdots & \ddots & \vdots \\ t & t & \cdots & s \end{bmatrix} = t\,\mathbf{1}\mathbf{1}^\top + (s-t) I_m,$$

where $\mathbf{1} \in \mathbb{R}^m$ is the all-ones vector. We know that $\mathbf{1}\mathbf{1}^\top$ is rank-1:

$$\mathbf{1}\mathbf{1}^\top \mathbf{1} = m\mathbf{1}, \qquad \mathbf{1}\mathbf{1}^\top v = 0 \quad \text{for all } v \perp \mathbf{1}.$$

Thus, $\mathbf{1}$ is an eigenvector of $\mathbf{1}\mathbf{1}^\top$ with eigenvalue m, and any vector orthogonal to $\mathbf{1}$ is an eigenvector with eigenvalue 0. Applying this to $X^\top X = t\,\mathbf{1}\mathbf{1}^\top + (s-t) I_m$ gives:

$$\begin{cases} \lambda_1 = (s-t) + tm = s + (m-1)t, & \text{(eigenvector: } \mathbf{1}\text{)}, \\ \lambda_2 = \cdots = \lambda_m = (s-t), & \text{(eigenspace: } \mathbf{1}^\perp\text{)}. \end{cases}$$

Hence, the singular values of X (the square roots of eigenvalues of $X^\top X$) are

$$\boxed{\sigma_1 = \sqrt{s + (m-1)t}, \qquad \sigma_2 = \cdots = \sigma_m = \sqrt{s-t}.}$$

(iii) [5 points] Let $\mathcal{C}_r$ be the set of all matrices obtained by selecting any r columns of X and setting the others to zero, and let X_r denote the rank-r truncated SVD of X as in part A(iv). Assume that each student can belong to multiple clubs, and that the clubs satisfy the *equal-size / constant-overlap* described in part B(ii). Show that for any $r < m$,

$$\|X - X_r\|_F \;<\; \min_{Y \in \mathcal{C}_r} \|X - Y\|_F,$$

i.e., *the rank-r SVD strictly outperforms any column-selection (club-removal) strategy when overlaps exist.*

From part (e), the club–club Gram matrix is

$$X^\top X = t\,\mathbf{1}\mathbf{1}^\top + (s-t)I_m,$$

which has eigenvalues

$$\lambda_1 = s + (m-1)t, \qquad \lambda_2 = \cdots = \lambda_m = s - t.$$

Hence, the singular values of X are

$$\sigma_1 = \sqrt{s + (m-1)t}, \qquad \sigma_2 = \cdots = \sigma_m = \sqrt{s-t}.$$

(i) Rank-r truncated SVD. The minimum error among all rank-r approximations is obtained by truncating the r largest singular values:

$$\|X - X_r\|_F^2 = \sum_{j=r+1}^{m} \sigma_j^2 = (m-r)(s-t).$$

Here, all dropped singular values are equal to $\sqrt{s-t}$ for $r \geq 1$.
(ii) Column-selection approximation. For any $Y \in \mathcal{C}_r$, the difference $X - Y$ contains exactly the $m - r$ columns that are zeroed out. Since each column of X has norm $\sqrt{s}$, the Frobenius norm separates across columns:

$$\|X - Y\|_F^2 = \sum_{j \notin S} \|X_{\cdot j}\|_2^2 = \sum_{j \notin S} s = (m-r)s,$$

independent of which r columns are kept. Therefore,

$$\min_{Y \in \mathcal{C}_r} \|X - Y\|_F^2 = (m-r)s.$$

(iii) Comparison. For any $t > 0$,

$$\|X - X_r\|_F^2 = (m-r)(s-t) \;<\; (m-r)s = \min_{Y \in \mathcal{C}_r} \|X - Y\|_F^2.$$

Thus,

$$\boxed{\|X - X_r\|_F \;<\; \min_{Y \in \mathcal{C}_r} \|X - Y\|_F.}$$

Interpretation. When $t = 0$ (disjoint clubs), the columns of X are orthogonal, so truncated SVD and column selection yield identical results. However, when $t > 0$, the club columns are no longer orthogonal due to member overlap. In this case, the SVD can exploit correlations between overlapping clubs by forming optimal linear combinations of columns, while simple column selection cannot. Hence, SVD strictly outperforms any "keep the r largest clubs" approach whenever overlaps exist.

5 [20 points]Frequent Itemsets

(a) [8 points] Prove or disprove the following.

(i) [2 points] If $X \subseteq X'$, then $\text{conf}(X' \to Y) \leq \text{conf}(X \to Y)$.

Let $X = \{A\}$, $X' = \{A, B\}$ (so $X \subseteq X'$), and $Y = \{C\}$. Consider the dataset of $N = 4$ *pairwise distinct* transactions:

T1	$\{A, B, C\}$
T2	$\{A, B, D\}$
T3	$\{A, D\}$
T4	$\{A\}$

$$\text{conf}(A \to C) = \frac{\text{support}(A \cup C)}{\text{support}(A)} = \frac{1}{4} = 0.25$$

$$\text{conf}(AB \to C) = \frac{\text{support}(AB \cup C)}{\text{support}(AB)} = \frac{1}{2} = 0.50.$$

(ii) [2 points] If $X \subseteq X'$, then $\text{interest}(X' \to Y) \leq \text{interest}(X \to Y)$.

$$\Pr(C) = \frac{\text{support}(C)}{N} = \frac{1}{4} = 0.25$$

$$\text{interest}(A \to C) = 0.25 - 0.25 = 0$$

$$\text{interest}(AB \to C) = 0.50 - 0.25 = 0.25.$$

(iii) [4 points] Let $\mathcal{D}$ be a dataset of transactions (possibly containing empty set), and let

$$I(\mathcal{D}) = \{\, a \mid \text{support}(\{a\}) > 0 \text{ in } \mathcal{D} \,\}.$$

There exists a dataset $\mathcal{D}$ consisting of *exactly* 16 *distinct transactions with no duplicates* such that

$$\text{interest}(A \to B) = 0 \quad \text{for } \textit{every } A, B \in I(\mathcal{D}),\ A \neq B\ .$$

Let $U = \{a, b, c, d\}$. Take the dataset $\mathcal{D}_{16} = \mathcal{P}(U)$, i.e., *all* $2^4 = 16$ subsets of U (including $\emptyset$), each appearing exactly once. Then $N = 16$ and for any distinct singletons $A, B \in U$,

$$\text{support}(\{A\}) = 8, \qquad \text{support}(\{B\}) = 8, \qquad \text{support}(\{A, B\}) = 4.$$

Hence

$$\text{conf}(A \to B) = \frac{\text{support}(\{A, B\})}{\text{support}(\{A\})} = \frac{4}{8} = \frac{1}{2} \quad \text{and} \quad \frac{\text{support}(\{B\})}{N} = \frac{8}{16} = \frac{1}{2},$$

so for every ordered singleton rule $A \to B$ we have

$$\text{interest}(A \to B) = \text{conf}(A \to B) - \frac{\text{support}(\{B\})}{N} = 0.$$

(b) [12 points] You have a market-basket dataset $\mathcal{D}$ with absolute support threshold s. Pick an arbitrary sample $S \subset \mathcal{D}$ by keeping each basket independently with probability p. Let

$$F \;:=\; \{\, X : \text{support}_D(X) \geq s \,\}$$

be the true frequent itemsets in $\mathcal{D}$. In this problem, we consider the Toivonen algorithm using 10 different support thresholds:

$$\tau_k \;:=\; \beta_k ps \quad \text{where } \beta_k = \frac{1}{10}k \quad (k = 1, 2, \ldots, 10)$$

Define

$$F_S^{(k)} \;:=\; \{\, X : \text{support}_S(X) \geq \tau_k \,\} \quad \text{for each } k \in \{1, \ldots, 10\}.$$

(i) [1 point] For the same, single sample S, show that the sets of frequent itemsets $F_S^{(k)}$ are nested as follows.

$$F_S^{(10)} \subseteq F_S^{(9)} \subseteq \cdots \subseteq F_S^{(1)}$$

Clear since $\tau_k < \tau_{k-1}$

(ii) [3 points] For the same, single sample S, decide whether

$$\text{NB}\big(F_S^{(10)}\big) \;\subseteq\; \text{NB}\big(F_S^{(9)}\big) \;\subseteq\; \cdots \;\subseteq\; \text{NB}\big(F_S^{(1)}\big)$$

must hold where $\text{NB}(F)$ denotes the negative border of F. Prove your claim or disprove it by giving an explicit counterexample.

No. Let $ps = 100$. The two relevant thresholds are:

* $\tau_{10} = \beta_{10} ps = \frac{10}{10} \times 100 = 100$
* $\tau_9 = \beta_9 ps = \frac{9}{10} \times 100 = 90$

Now, consider an itemset $X = \{a, b, c\}$ and a sample S with the following support counts:

* $\text{support}_S(\{a, b, c\}) = 95$
* $\text{support}_S(\{a, b\}) = 105$
* $\text{support}_S(\{a, c\}) = 105$
* $\text{support}_S(\{b, c\}) = 105$

Let's check if $X = \{a, b, c\}$ belongs to $\text{NB}(F_S^{(10)})$ and $\text{NB}(F_S^{(9)})$.

Analysis for $\text{NB}(F_S^{(10)})$:

1. Is $\{a, b, c\} \notin F_S^{(10)}$? Yes, because $\text{support}_S(\{a, b, c\}) = 95$, which is less than $\tau_{10} = 100$.
2. Are all immediate subsets of $\{a, b, c\}$ in $F_S^{(10)}$? Yes, because the supports of $\{a, b\}$, $\{a, c\}$, and $\{b, c\}$ are all 105, which is greater than or equal to $\tau_{10} = 100$.

Since both conditions are met, we conclude that $\{\mathbf{a}, \mathbf{b}, \mathbf{c}\} \in \text{NB}(\mathbf{F_S^{(10)}})$.

Analysis for $\text{NB}(F_S^{(9)})$:

1. Is $\{a, b, c\} \notin F_S^{(9)}$? No. The condition fails because $\text{support}_S(\{a, b, c\}) = 95$, which is greater than or equal to $\tau_9 = 90$. Therefore, $\{a, b, c\} \in F_S^{(9)}$.

(iii) [3 points] For the same, single sample S, assume *Strategy 1* uses only $k = 1$ ($\beta_1 = 0.1$), while *Strategy 2* uses only $k = 10$ ($\beta_{10} = 1.0$). Compare and contrast these two strategies with respect to two aspects: (1) the computational cost of Pass 2, and (2) the probability of success.

Conclusion: *Strategy 1* (conservative, $k = 1$) is expected to have a significantly higher computational cost than *Strategy 2* (aggressive, $k = 10$).

Justification: The computational cost of Pass 2 is primarily determined by the total number of candidate itemsets whose true support must be counted over the entire dataset $\mathcal{D}$. The set of candidates for a given strategy k is $C_k = F_S^{(k)} \cup \mathrm{NB}(F_S^{(k)})$.

From the proof in part (i), we know that a lower threshold results in a larger set of frequent itemsets. Since $\tau_1 \ll \tau_{10}$, we have the established relationship:

$$F_S^{(10)} \subseteq F_S^{(1)}$$

Typically, the size of $F_S^{(k)}$ is the dominant component of the total candidate set size. Strategy 1 uses a very low support threshold ($\tau_1 = 0.1ps$), which means a vast number of itemsets will be considered frequent in the sample. This makes the set $F_S^{(1)}$ extremely large.

In contrast, Strategy 2 uses a much stricter threshold ($\tau_{10} = ps$), which will result in a much smaller set of frequent itemsets, $F_S^{(10)}$. Even though $\mathrm{NB}(F_S^{(10)})$ might be larger than $\mathrm{NB}(F_S^{(1)})$, the enormous difference in the sizes of $F_S^{(1)}$ and $F_S^{(10)}$ means that $|C_1| \gg |C_{10}|$. Therefore, Strategy 1 will be far more computationally expensive in Pass 2.

Probability of Success

Conclusion: *Strategy 1* has a much higher probability of success. *Strategy 2* is highly vulnerable to failure due to a specific type of sampling error.

Justification: The Toivonen algorithm succeeds if and only if no true frequent itemset (from $\mathcal{D}$) is found in the negative border of the sample. Failure is often caused by **false negatives** from sampling. A false negative occurs when an itemset X that is truly frequent in $\mathcal{D}$ (i.e., $\mathrm{support}_{\mathcal{D}}(X) \geq s$) is found to be infrequent in the sample S by chance (i.e., $\mathrm{support}_S(X) < \tau_k$).

Vulnerability of Strategy 2 ($k = 10$): This strategy uses a sample threshold of $\tau_{10} = ps$. Consider an itemset X that is barely frequent in the full dataset, e.g., $\mathrm{support}_{\mathcal{D}}(X) = s$. The expected support for X in the sample is ps. Due to random statistical fluctuations, it is quite likely that its actual support in the sample, $\mathrm{support}_S(X)$, will be slightly less than the expectation (e.g., $ps - 1$). In this scenario:

* Since $\mathrm{support}_S(X) < ps = \tau_{10}$, X is deemed infrequent and $X \notin F_S^{(10)}$.
* If all immediate subsets of X are sufficiently frequent in the sample to be included in $F_S^{(10)}$, then by definition, X will be placed in the negative border, $\mathrm{NB}(F_S^{(10)})$.

Because a true frequent itemset X is now in the negative border, the algorithm fails for $k = 10$. This is why a $\beta < 1$ is generally recommended: to create a "safety margin" against such minor, but probable, statistical deviations.

Robustness of Strategy 1 ($k = 1$): This strategy uses a very low threshold of $\tau_1 = 0.1ps$. For the same true frequent itemset X to cause a failure, its

sample support would have to fall from an expected value of at least ps to a value below $0.1ps$. This represents a massive deviation from the statistical mean, which is an extremely rare event. Therefore, Strategy 1 provides a very large safety margin and is far more likely to succeed.

(iv) [5 points] Now, assume that we have drawn two independent samples, S_A and S_B, from the dataset $\mathcal{D}$ using the same probabilistic method described above.

First, for a given sample S, we define its *Success Index Set*, $K_{succ}(S)$, as follows. This set indicates which of the 10 thresholds lead to a successful run of the Toivonen algorithm.

$$K_{succ}(S) := \{k \in \{1, \ldots, 10\} \mid \forall X \in \mathrm{NB}(F_S^{(k)}), \mathrm{support}_{\mathcal{D}}(X) < s\}$$

In other words, $k \in K_{succ}(S)$ means that when using the threshold τ_k, its corresponding negative border is "safe," allowing the algorithm to terminate successfully.

For the two independently drawn samples S_A and S_B, decide whether the equality $K_{succ}(S_A) = K_{succ}(S_B)$ must always hold. Prove your claim, or disprove it by providing an explicit counterexample.

No. **1. Setup:** Let the global support threshold be $s = 50$ and the sampling probability be $p = 0.1$. Consider a specific itemset $X = \{a, b\}$ such that:

* X is a true frequent itemset: $\mathrm{support}_{\mathcal{D}}(X) = 50$.
* Its immediate subsets $\{a\}$ and $\{b\}$ are extremely frequent.

We will analyze the success for the threshold index $k = 6$, where $\beta_6 = 0.6$. The corresponding sample threshold is $\tau_6 = \beta_6 ps = 0.6 \times 0.1 \times 50 = 3$.

2. Independent Samples: Due to sampling variability, the following two outcomes for S_A and S_B are possible:

* **Sample S_A (Statistically fortunate):** The support of X is at or above its expectation ($ps = 5$). Let $\mathrm{support}_{S_A}(X) = 5$.
* **Sample S_B (Statistically unfortunate):** The support of X is below the threshold τ_6. Let $\mathrm{support}_{S_B}(X) = 2$.

In both samples, assume the subsets $\{a\}$ and $\{b\}$ remain highly frequent.

3. Analysis of Success for $k = 6$:

* **For sample S_A:** Since $\mathrm{support}_{S_A}(X) = 5 \geq \tau_6$, the itemset X is considered frequent in the sample ($X \in F_{S_A}^{(6)}$). By definition, X cannot be in the negative border $\mathrm{NB}(F_{S_A}^{(6)})$. Assuming no other true frequent itemsets land in the negative border, the algorithm succeeds for this threshold. Thus, $\mathbf{6} \in \mathbf{K_{succ}(S_A)}$.
* **For sample S_B:** Here, $\mathrm{support}_{S_B}(X) = 2 < \tau_6$, so X is infrequent in the sample ($X \notin F_{S_B}^{(6)}$). Since its subsets are frequent, X meets the conditions to be in the negative border: $\mathbf{X} \in \mathrm{NB}(\mathbf{F_{S_B}^{(6)}})$. However, X is a true frequent itemset ($\mathrm{support}_{\mathcal{D}}(X) = 50 \not< s$). This violates the condition for success. The algorithm fails for this threshold. Thus, $\mathbf{6} \notin \mathbf{K_{succ}(S_B)}$.

6 [25 points] MapReduce and Spark

A quantile is the element at a specific position in an ordered dataset, such as the median. For a dataset $X = \{x_1, \ldots, x_N\}$ of distinct values, define the rank of an element x as the number of elements which is less than or equal to x: $\texttt{rank}_X(x) = |\{x_i \in X : x_i \leq x\}|$. We say x is an ϕ-quantile of X if $x \in X$ and $\texttt{rank}_X(x) = \lceil \phi N \rceil$ where $0 < \phi \leq 1$. Here, $\lceil a \rceil$ returns the smallest integer greater than or equal to a.

(a) [1 point] Compute the 0.5-quantile of $X = \{11, 21, 24, 61, 81, 39, 89, 56, 12, 51\}$.

Answer. 39

Criterion.

- No partial points.

(b) [5 points] We want to compute a quantile over a large collection of numbers in a distributed way. This can be achieved with a *decomposable* function; a function f over a set X is said to be *decomposable* if there exists a function g such that $f(X) = g(f(X_1), f(X_2))$ for all disjoint subsets X_1, X_2 with $X = X_1 \cup X_2$ and $X_1 \cap X_2 = \emptyset$. Show that the 0.5-quantile of a set X is **NOT** decomposable; provide a specific counterexample.

Hint: Find X, Y with partitions $X = X_1 \cup X_2$, $Y = Y_1 \cup Y_2$ such that $f(X) \neq f(Y)$ but $f(X_1) = f(Y_1)$ and $f(X_2) = f(Y_2)$, thereby showing that no such g can exist.

Answer. $X_1 = \{0, 2, 4\}, X_2 = \{1, 3, 5\}, Y_1 = \{1.5, 2, 2.5\}, Y_2 = \{2.4, 3, 3.5\}$

Criterion.

- (1 point) Express X_1, X_2, Y_1, Y_2 with proper X, Y.
- (2 points) Show $f(X_1) = f(Y_1), f(X_2) = f(Y_2)$.
- (2 points) Show $f(X) \neq f(Y)$.
- deduct 1 point for minor mistake.
- If you don't use **Hint**, then you will get points based on clarity and logical completeness.

(c) [4 points] To make the quantile computation decomposable, we define *ϵ-approximate ϕ-quantile* of X as a value $x \in X$, where $\epsilon \ll 1$ is small and $|X| = N$, satisfying

$$|\texttt{rank}_X(x) - \phi N| \leq \epsilon N.$$

This allows an error of at most ϵN compared to the ϕ-quantile of X. Let X be as defined in (a). **Choose five elements in** X that cover all 0.1-approximate ϕ-quantiles of X for $\phi \in (0, 1]$. (That is, there exists a query function $q : (0, 1] \to \hat{X}$ returning the 0.1-approximate ϕ-quantile of X where $\hat{X} \subset X$ with $|\hat{X}| = 5$.)

Answer. 11, 21, 39, 56, 81

Criterion.

- deduct 1 point for each incorrect or missing element.

(d) [5 points] As in (c), all ϵ-approximate quantiles can be answered using a subset of elements. We define an *ϵ-approximate quantile summary* of X as

$$S := \{s_1, \cdots, s_M\} \subset X$$

where there exist functions $\texttt{rmin}_X, \texttt{rmax}_X : S \to \mathbb{N}$ such that $s_1 \leq \cdots \leq s_M$, and

- $\texttt{rmin}_X(s_i) \leq \texttt{rank}_X(s_i) \leq \texttt{rmax}_X(s_i)$ for all $1 \leq i \leq M$.
- $\texttt{rmax}_X(s_{i+1}) - \texttt{rmin}_X(s_i) \leq 2\epsilon|X|$ for all $1 \leq i < M$.

By bounding the $\texttt{rank}_X$ between $\texttt{rmin}_X$ and $\texttt{rmax}_X$, each element s_i covers a range of quantiles in X. Let the summary function $S(\cdot)$ map a dataset X to its ϵ-approximate quantile summary S. Then, this summary function is decomposable: there exists a combine function c such that $S(X) = c(S(X_1), S(X_2))$ for all disjoint subsets X_1, X_2 with $X = X_1 \cup X_2$ and $X_1 \cap X_2 = \emptyset$. Design such a function c; no proof is required.

Hint: Let $S(X_1) = \{p_1, \cdots, p_{M_1}\} \subset X_1$ and $S(X_2) = \{q_1, \cdots, q_{M_2}\} \subset X_2$. Then design $S(X) = \{s_1, \cdots, s_M\} \subset X$ with $M = M_1 + M_2$, by defining $\texttt{rmin}_X$ and $\texttt{rmax}_X$ in terms of $\texttt{rmin}_{X_1}, \texttt{rmax}_{X_1}, \texttt{rmin}_{X_2}$, and $\texttt{rmax}_{X_2}$.

Answer. $S(X) = S(X_1) \cup S(X_2)$. For $s \in S(X)$, let X_i be the subset that contains s and X_j be the other subset, with corresponding summaries S_i and S_j. Let p be the largest element in S_j which is smaller than s, and n be the smallest element in S_j which is larger than s. Then:

$$\texttt{rmin}_X(s) = \texttt{rmin}_{X_i}(s) + \begin{cases} \texttt{rmin}_{X_j}(p), & \text{if } p \text{ exists,} \\ 0, & \text{otherwise,} \end{cases}$$

$$\texttt{rmax}_X(s) = \texttt{rmax}_{X_i}(s) + \begin{cases} \texttt{rmax}_{X_j}(n) - 1, & \text{if } n \text{ exists,} \\ \texttt{rmax}_{X_j}(\max S_j), & \text{otherwise,} \end{cases}$$

Criterion.

- (1 point) Show $S(X) = S(X_1) \cup S(X_2)$.
- (2 points) Correctly define $\texttt{rmin}_X$ in terms of $\texttt{rmin}_{X_1}, \texttt{rmin}_{X_2}$.
- (2 points) Correctly define $\texttt{rmax}_X$ in terms of $\texttt{rmax}_{X_1}, \texttt{rmax}_{X_2}$.
- deduct 1 point for minor mistake.
- If you don't use **Hint**, then you will get points based on clarity and logical completeness.

In Spark, the ϵ-approximate ϕ-quantile can be computed in a distributed manner:

```
S = sc.parallelize(X) \
    .mapPartitions(lambda Xi: summary(Xi, eps)) \
    .treeReduce(lambda Sj, Sk: merge(Sj, Sk, eps))

result = query(S, phi, eps)          # problem (g)
```

Here, `mapPartitions` creates a summary S_i for each partition $X_i \subset X$, and `treeReduce` recursively merges S_j and S_k into a new summary S_l. The functions are defined as follows. Note that `floor(a)` returns the greatest integer less than or equal to `a`.

```
def summary(Xi, eps):
    Ni = len(Xi); Si = []
    for r, x in enumerate(sorted(Xi)):
        t = {'value': x, 'rmin': r+1, 'rmax': r+1}
        if (r % floor(2*eps*Ni) == 0) or (r == Ni - 1):
            Si.append(t)
    return Si

def merge(Sj, Sk, eps):
    Nl = Sj[-1]['rmax'] + Sk[-1]['rmax']
    Sl = combine(Sj, Sk)            # problem (e)
    Sl = compress(Sl, Nl, eps)      # problem (f)
    return Sl
```

(e) [6 points] In the Map stage, `summary` generates an ϵ-approximate quantile summary S_i for each partition X_i. In the Reduce stage, given summaries S_j and S_k for X_j and X_k, respectively, `combine` should produce an ϵ-approximate quantile summary S_l for $X_j \cup X_k$. Fill (1) and (2) in the code below to satisfy this property.

```
def combine(Sj, Sk):
    Z = sorted(Sj + Sk, key=lambda s:  s['value']); Sl = []
    for z in Z:
        Sz, Sn = (Sj, Sk) if z in Sj else (Sk, Sj)
        prev = max([t for t in Sn if t['value'] < z['value']],
                key=lambda t:  t['value'], default=None)
        next = min([t for t in Sn if t['value'] > z['value']],
                key=lambda t:  t['value'], default=None)
        rmin = [                    (1)                    ]
        rmax = [                    (2)                    ]
        t = {'value':z['value'], 'rmin':rmin, 'rmax':rmax}
        Sl.append(t)
    return Sl
```

Answer.

(1) `rmin = z['rmin'] + (prev['rmin'] if prev is not None else 0)`

(2) `rmax = z['rmax'] + ((next['rmax'] - 1) if next is not None else Sn[-1]['rmax'])`

Criterion.

- 3 points for each item.
- deduct 1 point for minor mistake.

(f) [2 points] In `compress`, we make the summary compact by ensuring that

$$\texttt{rmax}_X(v_{i+2}) - \texttt{rmin}_X(v_i) > 2\epsilon|X|$$

for all $1 \leq i < s - 1$. Fill (3) in the code below so that this condition is satisfied.

```
def compress(S, N, eps):
    R = [S[0]]; prev = S[0]; T = 2*eps*N
    for t in S[1:]:
        if [                (3)                ]:
            R.append(prev)
        prev = t
    R.append(S[-1])
    return R
```

Answer. `t['rmax'] - R[-1]['rmin'] > T`

Criterion.

- deduct 1 point for minor mistake.

(g) [2 points] We can design `query(S, phi, eps)` that returns an ϵ-approximate ϕ-quantile of X, given an ϵ-approximate quantile summary of X. Fill (4) in the code below.

```
def query(S, phi, eps):
    N = S[-1]['rmax']; target = phi*N; tol = eps*N
    for i in range(len(S)):
        if [                (4)                ]:
            return S[i]['value']
    return S[-1]['value']
```

Answer. `S[i]['rmin'] - tol <= target <= S[i]['rmax'] + tol`

Criterion.

- deduct 1 point for minor mistake.